RASHTRIYA ISPAT NIGAM LIMITED –

VISAKHAPATNAM STEEL PLANT

PROJECT REPORT 0N

STUDY OF POWER DISTRIBUTION IN VSP

Dissertation Report Submitted in fulfilment for the award of certificate of Internship Training in Electrical Engineering from RINL – VSP

Submitted To:

Department Of

DISTRIBUTION NETWORK (DNW)

Submitted By:

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INDIAN INSTITUTE OF TECHNOLOGY -GUWAHATI

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CERTIFICATE

This is to certify that the work embodied in this project, Dissertation Report entitled as “STUDY OF POWER DISTRIBUTION IN VSP” being submitted by KUNDRAPU MADHU (Trainee no - 100052166) in complete fulfilment of the requirement for the award of “Internship Training” in Electrical Engineering discipline to RashtriyaIspatNigam Limited – Visakhapatnam Steel Plant, Visakhapatnam (A.P) during the year 2026 is a record of Bonified Piece of work, Carried out under my supervision and the guidance in the Department of Electrical Engineering, Distribution Network RINL – VSP .

Approved By

N MOHAN RAO

SR. Manager DNW , RINL – VSP

DECLARATION

I hereby declare that this Project Report entitled "A STUDY O OF POWER DISTRIBUTUION in RASHTRIYA ISPAT NIGAM LIMITED (RINL) -VSP , submitted in the complete fulfilment of the requirements for the award of “Internship Training” in Electrical Engineering discipline to RashtriyaIspat Nigam Limited – Visakhapatnam Steel Plant , Visakhapatnam (A.P),is an authentic record of my own work.

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ACKNOWLEDGMENT

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Steel Plant, for his assistance in completion of the project

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CONTENTS

INTRODUCTIONTOVSP

Steel comprises one of the most important inputs in all sectors of economy. Steel industry is both a basic and a core industry. The economy of any nation depends on a strong base of iron and steel industry in that country. History has shown that countries having strong potentiality of iron and steel production have played a prominent role in the advancement of civilization in the world. Steel is such a versatile commodity that every object we see in our day-to-day life has used steel either directly or indirectly. The per capita consumption of steel in India during 1970's was around 10Kg compared to about 700Kg obtaining in many advanced countries, over 800Kg in Japan, which was considered to be very low. Viewing in the backdrop of Indian populations which was standing at about 1.21 billion presently, even 10Kg of increase in steel consumption would need the setting up of an additional production of a billion tons of steel per year. Keeping this in view the government of India had cleared the decision of setting up of a shore based integrated steel plant at Visakhapatnam in the year 1970 as part of its vision

FIG 1 : VIZAG STEEL PLANT

a) BACKGROUND

The decision of government of India to set up an integrated steel plant at Visakhapatnam was announced by the Prime Minister Smt.Indira Gandhi in the parliament on 17th April 1970. Visakhapatnam Steel Plant (VSP), a public sector undertaking, is under Rashtriya
Ispat Nigam Limited. It is one of the most sophisticated modern plants. The foundation stone was laid by the former Prime Minister Smt. Indira Gandhi in the year 1972. It is strategically located on the coast of Bay of Bengal in the state of Andhra Pradesh. It is an integrated steel plant with the state-of-the-art technology. Overcoming the perils of recession in the steel market it has turned into a profit-making organization with proper strategic decisions. A professionally organized steel plant, VSP has been the recipient of the ISO 9001 (for Quality), OHSAS 18001(for Occupational Health & Safety), and the ISO 14001 (for Environmental Management) and 5s certifications. Armed with these three international standards and an advocate to the TQM, it has achieved the six-sigma level of efficiency. The management of the plant is organized into production, marketing, finance, personnel, purchase and other auxiliary departments. The plant is designed to produce 3.0 million tones of liquid steel. It possesses a strong well manpower of 17,000 employees. The organization is house of technology where international levels of efficiency are being pursued in terms of productivity and specific energy consumption. For the financial year 2003-04 the organization recorded a turnover of Rs. 6174 crores.

VSP has achieved a turn around to post a net profit of nearly Rs. 1521 crores. In addition to a wide range of steel products there are other by- products, which are produced by VSP like tar, pitch and the noted fertilizer "pushkala". VSP because of its geographical advantage and standard products has carved a niche for itself. Nearly 40% of the South Indian domestic market has been captured for the steel products. International customers are from the countries of China, Singapore, Russia, Nepal, Sri Lanka, USA, Japan, UAE etc.

b) MAJOR DEPARTMENTS IN V.S.P

> Raw Materials Handling Plant (RMHP)

> Coke Ovens & Coal Chemicals Plant (CO & CCP)

> Sinter Plant (SP)

> Blast Furnace (BF)

> Steel Melt Shop (SMS)

> Air Separation Plant (ASP)

> Rolling Mills

- Light & Medium Merchant Mill (LMMM)
- Wire Rod Mill (WRM)
- Medium Merchant & Structural Mill (MMSM)
- Roll Shop & Repair Shop (RS & RS)

> Calcining and Refractory Material Plant (CRMP)

POWER DISTRIBUTION IN VSP

a) DEFINITION OF POWER DISTRIBUTION There are different sources from which the power can be produced for running the machines in industry. The chief source is sun, but its energy cannot be efficiently utilized. The other sources from which the power can be obtained are:

- Fuel (coal & oil)
- Water (hydel)
- From nuclear bombardment (uranium)

The electricity produced at the generating station is transmitted to the consumers through a network of transmission and distribution systems. It is often difficult to draw a line between the transmission and distribution systems of a large power system. It is impossible to distinguish the two merely by their voltage because what was considered as a high voltage a few years ago is now considered as a low voltage. In general, distribution system is the major and important part of a power system which distributes power to the consumers for utilization. The transmission and distribution systems are similar to man's circulatory system. The distribution system is compared with capillaries. They serve the same purpose of supplying the ultimate consumer in the city with the life giving blood of civilization-electricity.

"THE PART OF POWER SYSTEM WHICH DISTRIBUTES ELECTRIC POWER FOR LOCAL USE IS KNOWN AS DISTRIBUTION SYSTEM"

FIG2: Transmission towers

VOLTAGE LEVELS AT V.S.P

> Extra high voltage: 220KV
> High Voltage: 33/11/6.6KV
> Medium Voltage: 440 Volts
> Low Voltage: 230 Volts

- SOURCES OF POWER

Power requirement of V.S.P is met through captive generation as well as supply from A.P TRANSCO grid. The captive capacity of 270 MW is sufficient to meet all the plant needs in normal operation time. In case of outage of captive generation capacity due to breakdown, shutdown or other reasons, the short fall of power is availed from A.P TRANSCO grid.

- POWER REQUIREMENT

Integrated steel plants are major consumers of electricity, which specific consumption of power at around 600-650KWH/ton of liquid steel. The estimated annual power requirement of V.S.P, at full level of production in each department is 1932 Million KWH, which corresponds to average demand of 221 MW.

DISTRIBUTION NETWORKS(DNW)

Once the power is generated the next phase is its distribution to various load centres in optimal ways. At VSP the power distribution process is governed by a separate department known as DNW (DISTRIBUTION NETWORK). The steel plant generation capacity is 270MW is sufficient to meet the load demand.

d) MAJOR FUNCTIONS OF DNW DEPARTMENT

> Power system management with A.P TRANSCO & T.P.P

>Operation, inspection and maintenance of EHT and HT equipment's of plant

> Ensuring reliable protective system by periodic testing of relays and equipment's.

> Monitoring shop power consumption through SCADA network for optimization by shops

>Maintenance of PF and harmonic compensation for reactive power regulation.

The distribution network is classified into 6 zones:

- MRS (MAIN RECEIVING STATION) & LBSS -5(TPP
- AREA)

- LBSS 1,2,3,4
- RELAY TESTING GROUP(RTG)
- CABLE GROUP
- STORES
- MATERIAL PLANNING AND CONTRACTS (MPC)

MRS (MAIN RECEIVING STATION)

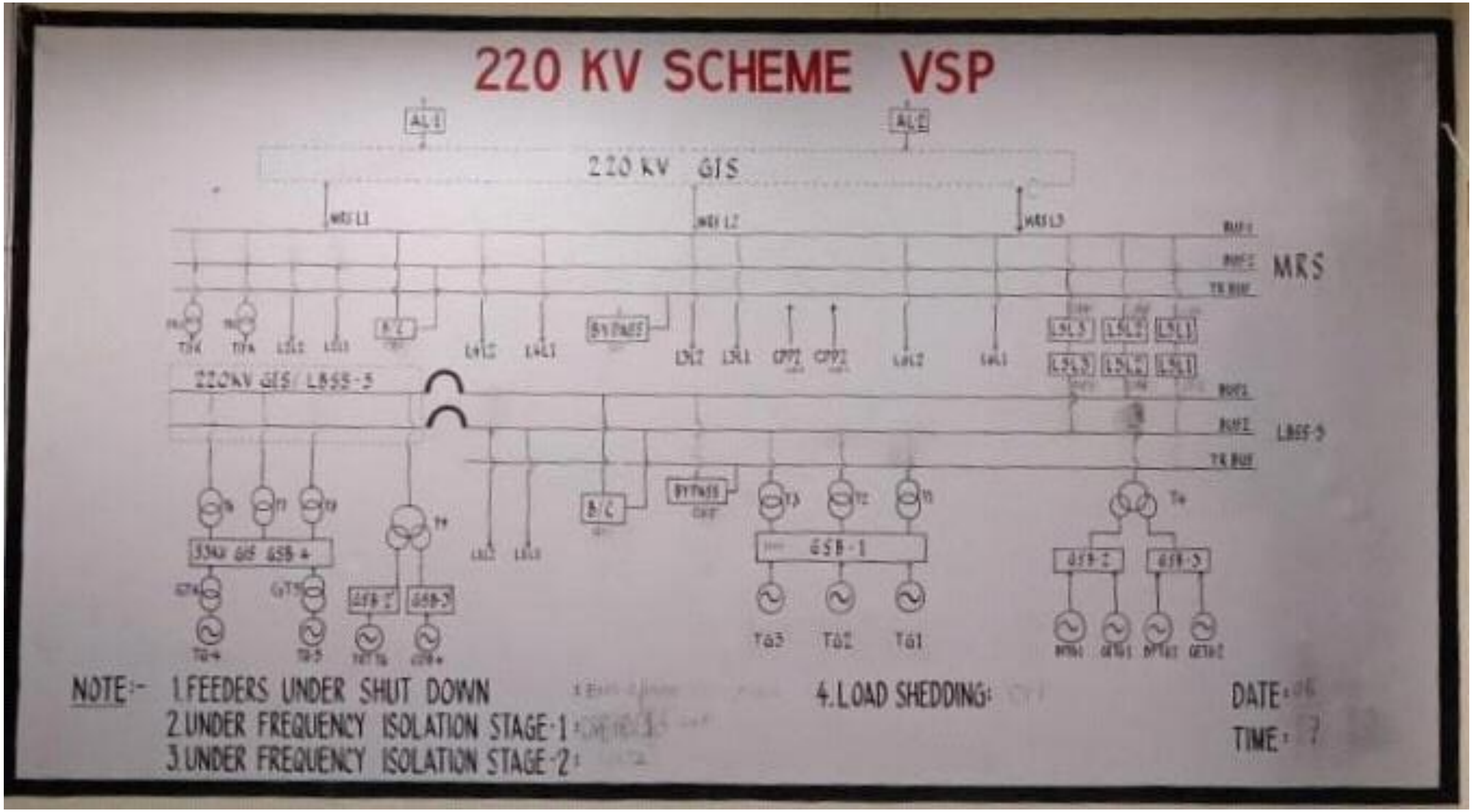
The power for the Steel Plant is supplied from APTRANSCO as well as from in plant generation. MRS is connected to APTRANSCO via double circuit 220KV transmissions line. It is also connected to TPP via three "TIE" lines named as TIE1,2,3.

There are three BUSES namely BUS-1, BUS-2 and TRANSFER BUS(T/R) are installed for supplying power to various loads. Different loads are connected to either BUS-1 or BUS-2. T/R BUS is used when the circuit breaker trips at the load side feeder trips. T/R BUS is can be charged by closing the isolater of BY-PASS circuit arrangement.

The MRS supplies power at 220KV through two overhead transmission lines to LBSS 2, LBSS-3, LBSS-4 and also to the township through CPRS by stepping down from 220KV to 33KV and 33KV to 11KV.

FIG 3: MRS Yard

In MRS there are altogether 21 bays connected to BUS-1 and BUS-2. The description of all the BAYS are as follows:


220 KV SCHEME VSP
AL1
AL2
220 KV GIS
MRS
220KV GIS/ LBSS-5
LBSS-5
BYPASS
B/C
GSB-1
TG3
TG2
TG1
GSB-2
GSB-3
NOTE:- 1.FEEDERS UNDER SHUT DOWN
2.UNDER FREQUENCY ISOLATION STAGE-1:
3.UNDER FREQUENCY ISOLATION STAGE-2:
4.LOAD SHEDDING:
DATE:
TIME:

SCADA:

The 220 KV, 11 KV and 6.6 KV distribution systems is monitored by a centralized Supervisory Control and Data Acquisition system. The plant generation, import/export and consumption in each unit are monitored through SCADA.

220 KV DISTRIBUTION NETWORK

Power is supplied to V.S.P from APTRANSCO switching station over 220KV lines on double circuit towers. Power is received at the Main Receiving Station (MRS) which is further distributed to various units within the plant. The entire plant is configured as five electrical load block step-down substations (designated as L.B.S.S-1 to 5) with 220KV transformers to step down power to 33/11/66KV for further distribution. MRS & LBSS-5 are interconnected by three tie lines for bi-directional power flow. LBSS 1 is connected to LBSS 5 by two radial lines. LBSS -2 3. 4 are connected to MRS by two radial lines each. MRS & LBSS 5 is having double bus and transfer bus arrangement. Two 220/33KV, 12.5 MVA transformers installed at MRS to feed townships through CPRS. Six 220/11/11KV, 80 MVA transformers are installed at LBSS 3. 4 and six 220/11 & 6.6 KV. 80 MVA and one. 220/33 KV transformers e installed at LBSS-2 & LBSS-1. LBSS-5 consist of five transformers in which three transformers has a rating of 11/220 KV, 63 MVA each. Transformer four has a rating of 11/11/220 KV, 50 MVA and transformer five has a rating of 11/220 KV, 90MVA. TPP consist of four turbo Generators in which three are having a capacity of 60MW each and one is having a capacity of 67.5MW. It also consists four auxiliary generators in which two are Back Pressure Turbo Generators (BPTG - 1 & BPTG -2) are having a capacity of 7.5 MW each and two are Gas Expansion Turbo Generators (GETG-1 & GETG-2) are having a capacity of 12 MW each. Turbo Generators having a capacity of 60 MW feeds Generator Switch Board (GSB-1) & auxiliary generators feeds Generator Switch Board (GSB-2 & GSB - 3) and turbo Generator having a capacity of 67.5 MW is connected to LBSS-5. Sum of the power generated by TG-1 & TG-3 is 120 MW, out of which 60MW directed to GSB - 1 and the rest 60 MW isa routed to LBSS 5. same way the total power generated by auxiliary generators (BPTG-1. BPTG-2, GETG- 1 & GETG-2) is 39 MW, out of which 19.5 MW are directed to GSB - 2 and GSB-3 and the rest to LBSS-5. Some amount of the power from LBSS-5 is routed to LBSS - 1 and the surplus power which still exists is routed to MRS, which in turn exports to the APTRANSCO grid.

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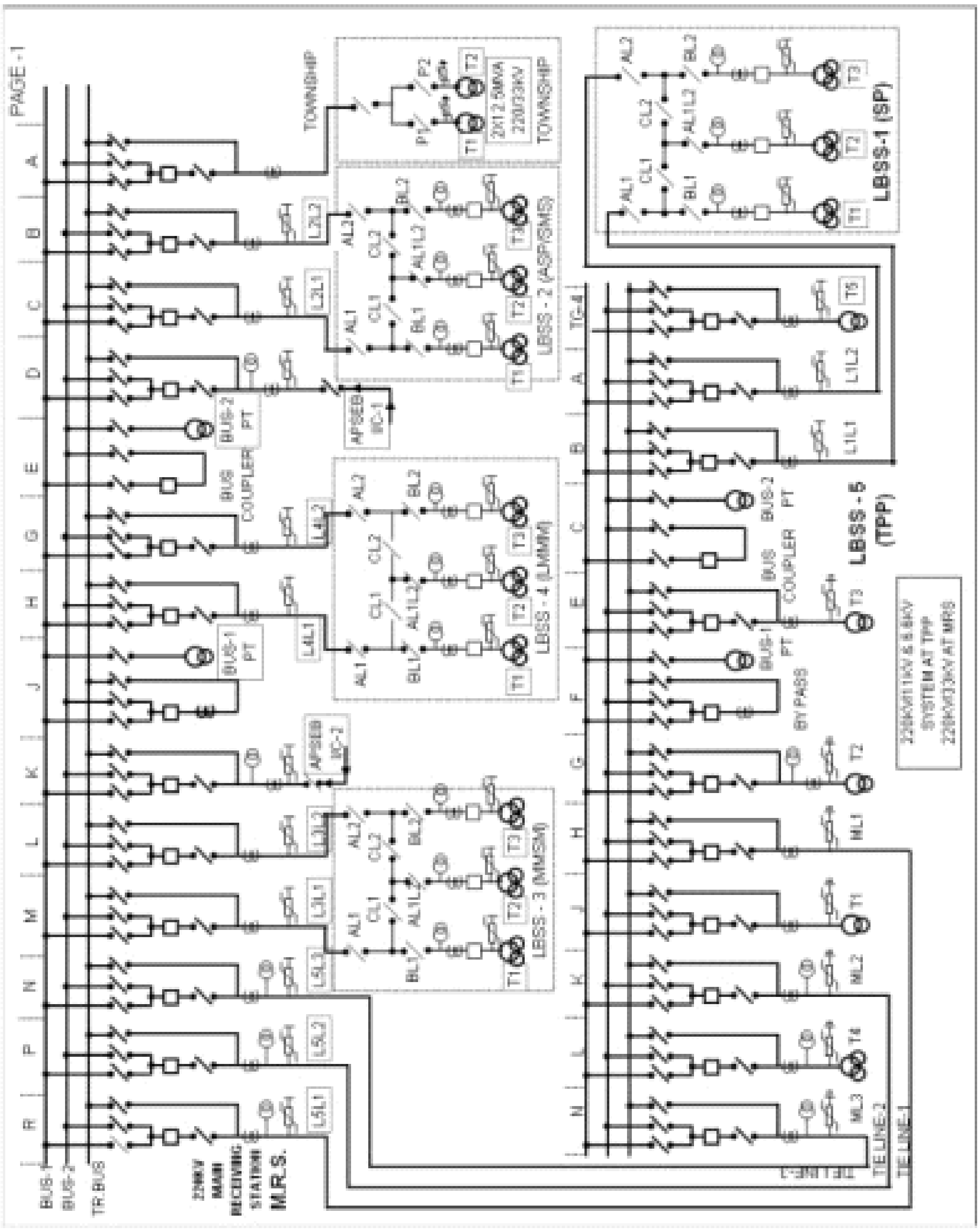


Single Line Diagram 220kV Power System of VSP

APSEBSUPPLYNETWORK

The Power Grid Corporation’s Sub-station adjacent to Ukkunagaram is connected to Vijayawada by a 400 kV line. It is also being connected to Jeypore, Orissa (Eastern Grid) through DC back-to-back arrangement of 500 MW capacity and by 400 kV AC double circuit line. Power is stepped down through a 315 MVA, 400/220 kV auto transformer at Power Grid Corporation Sub-station and is fed to the adjacent APSEB switching station. This switching station is also connected to Bommuru and Gajuwaka sub-stations by 220 kV double circuit lines. Bommuru sub-station is connected to generating stations at Vijayawada, Lower Sileru, Vijjeswaram, Kakinada and Jegurupadu. Gajuwaka sub station is connected to Upper Sileru. Two 1000 MW Thermal Power Stations areexpected to come up in the next few years at Visakhapatnam and close to steel plant. Power is supplied to VSP from APSEB switching station over two 220 kV lines on double circuit towers. Power is received at the Main Receiving Station (MRS) located near Main gate and further distributed to various units within the plant.

SOURCES OF POWER

Power requirement of VSP is met through captive generation as well as supply from APSEB grid. The captive capacity of 296.5 MW is sufficient to meet all the plant needs in normal operation time. In case of partial outage of captive generation capacity due to breakdown, shutdown or other reasons, the short fall of power is availed from ABSEB grid. Turbo Generators of VSP normally operate in parallel with state grid. Excess generation over and above plant load is exported to APSEB.The agreement with APSEB provides for a contract demand of 100 MVA and permit export of power. Tariff for import, export, demand charges, penalties etc. are stipulated. For purpose of billing, import and export energy is separately metered at Main Receiving Station.

UNIT COST OF POWER

- Generation cost : RS 2.50
- Import energy : RS 2.96
- Import demand per kva :RS.250
- Export energy :RS 1.76

POWER REQUIREMENT

Integrated Steel Plants are major consumers of electricity, with specific consumption of power at around 600-650 kWh/Ton of liquid steel. The estimated annual power requirement of Visakhapatnam Steel Plant, at full level of production in each shop (corresponding to 3.0 MT of liquid steel), is 1932 million kWh. This corresponds to an average demand of 221 MW. The estimated energy consumption and average demand of major shops is given below

SHOP	Annual Energy (10^6kW Hrs.)	Average Demand (MW)
RMHP	35	4.0
CO & CCP	171	19.5
SINTER PLANT	254	29.0
BLAST FURNACE	210	24.0
SMS & CCM	126	14.5
LMMM	100	11.5
WRM	118	13.5
MMSM	100	11.5
CRMP	35	4.0
TPP	310	35.0
ASP	258	29.5
COM. STATION & CWP	131	15.0
AUXILIARY SHOPS	20	2.5
WATER SUPPLY	15	2.0
TRAFFIC & OTHERS	7	1.0
TOWNSHIP	28	3.0
LOSSES	14	1.5
TOTAL	**1932**	**221.0**

However, production in VSP has been continuing beyond the rated capacity and

accordingly, the power requirement has also changed.

Import/export of power at vsp

- Import contract maximum demand :100 MVA
- Export allowed upto 25 MWCritical loads of vsp
- Co gas blower and MBC blowers.

- Flushing liquor and feed water pumps
- SMS lance lifting drives
- Co,BF,SMS,MILLS,TPP pump houses
- Circulating water system for LF
- Intake and make-up water pump houses
- Power plant auxiliaries
- Mills control systems
- Mills ECS and DMDA pumps
- Gas boosting

  stations Liquid

- nitrogen pumps

- Hospital.

Typical arrangement of power distribution at VSP

The further distribution of power from main receiving station is shown simply in the below line diagram

Main receiving station

|

Load block stepdown substation

|

LOAD BLOCK DISTRIBUTION SUBSTATION

HVLC	HVMCC	LCSS

HT LOADS

LT LOADS

220KVDISTRIBUTIONNETWORK

Power is supplied to V.S.P from APTRANSCO switching station over 220KV lines on double circuit towers. Power is received at the Main Receiving Station (MRS) which is further distributed to various units within the plant. The entire plant is configured as six electrical load block step-down substations (designated as L.B.S.S-1 to 6) with 220KV transformers to step-down power to 33/11/66KV for further distribution.

MRS & LBSS-5 are interconnected by three tie lines for bi-directional power flow.

LBSS -1 is connected to LBSS 5 by two radial lines.

LBSS -2, 3, 4 are connected to MRS by two radial lines each. MRS & LBSS 5 is having double bus and transfer bus arrangement. Two 220/33KV, 12.5 MVA transformers installed at MRS to feed townships through CPRS. Six 220/11/11KV, 80 MVA transformers are installed at LBSS 3, 4 and six 220/11 & 6.6 KV, 80 MVA and one 220/33 KV transformers are installed at LBSS-2 & LBSS - 1.

LBSS-5 consist of five transformers in which three transformers has a rating of 11/220 KV, 63 MVA each. Transformer four has a rating of 11/11/220 KV, 50 MVA and transformer five has a rating of 11/220 KV, 90MVA. TPP consist of four turbo Generators in which three are having a capacity of 60MW each and one is having a capacity of 67.5MW. It also consists of four auxiliary generators in which two are Back Pressure Turbo Generators (BPTG - 1 & BPTG - 2) are having a capacity of 7.5 MW each and two are

Gas Expansion Turbo Generators (GETG-1 & GETG-2) are having a capacity of 12 MW each.

FIG 4 : Transformer 220Kv/11Kv

Turbo Generators having a capacity of 60 MW feeds Generator Switch Board (GSB
1) & auxiliary generators feeds Generator Switch Board (GSB-2 & GSB-3) and turbo

Generator having a capacity of 67.5 MW is connected to LBSS-5. Sum of the power generated by TG-1 & TG-3 is 120 MW, out of which 60MW directed to GSB – 1 and the rest 60 MW is routed to LBSS -5, same way the total power generated by auxiliary generators (BPTG - 1, BPTG-2, GETG-1 & GETG-2) is 39 MW, out of which 19.5 MW are directed to GSB-2 and GSB-3 and the rest to LBSS-5. Some amount of the power from LBSS-5 is routed to LBSS - 1 and the surplus power which still exists is routed to MRS, which in turn exports to the APTRANSCO grid. Power for the township is imported from MRS which was initially exported from LBSS - 5. In case of any shortage, the MRS will import required power from APTRANSCO grid to cater for the requirements of township.

FIG 5 : TURBO GE

NERATOR

220KV FEEDERS at MRS YARD at VSP

BAY	- Description
A -Bay	- Township
B -Bay	- L2L2 LBSS2 Feeder2
C -Bay	- L2L1 LBSS2 Feeder1
D -Bay	- AL-1 AP Transco Line1
E -Bay	- Bus Coupler
G -Bay	- L4L2 LBSS4 Feeder2H
-Bay	- L4L1 LBSS4 Feeder1
J -Bay	- By Pass Bus
K -Bay	- AL-2 AP Transco Line2
L -Bay	- L3L2 LBSS3 Feeder2
M -Bay	- L3L1 LBSS3 Feeder1
Bay 14	- L6L2 LBSS6 Feeder2
Bay 15	- L6L1 LBSS6 Feeder2
Bay 18	- Tie-3 Tie Line 3 to LBSS5Bay
20	- Tie-2 Tie Line 2 to LBSS5Bay
21	- Tie-1 Tie Line 1 to LBSS5

LBSS (LOAD BLOCK STEPDOWN SUBSTATION)

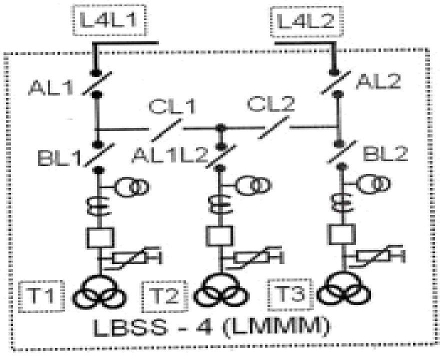
All together there are five LBSS-1,2,3,4&5. In extinction there are LBSS-6 and LBSS-7 also. All the LBSS are connected to different buses for equal distribution of loads. LBSS-3,LBSS-4 and Township are connected to BUS-1 and LBSS-2 is A connected to BUS-2. LBSS-1 via connected at LBSS-5. The power at LBSS are stepdown at the levels of 11 or 66KV depending upon the need. There are six numbers of switch boards at each LBSS. The power from LBSS is distributed to different to different LCSS(LOAD CENTRE STEP DOWN SUBSTATION) at same levels. At LCSS voltage is again stepdown to 415 or 220v as per the requirement. Each LBSS feeds power to different departments as shown in detail:

STATION DESIGNATION	AREAS COVERED

LBSS-1 (220/11/6.6 KV)	COCCP, RMHP, SP-1
LBSS-2 (220/11/6.6 KV) (220/33 KV)	ASP, BF-1&2, SMS-1, CRMP
LBSS-3 (220/11/11 KV)	MMSM
LBSS-4 (220/11/11 KV)	LMMM, WRM-1, PUMP-HOUSE 1&2
LBSS-5 (220/11 &220/11/11KV)	TPP, Plant essential category loads, KBR, Township & Hospitals
LBSS-6 & 7 (220/11/6.6 KV) (220/33 KV)	BF-3, SMS-2, WRM-2, SP-2

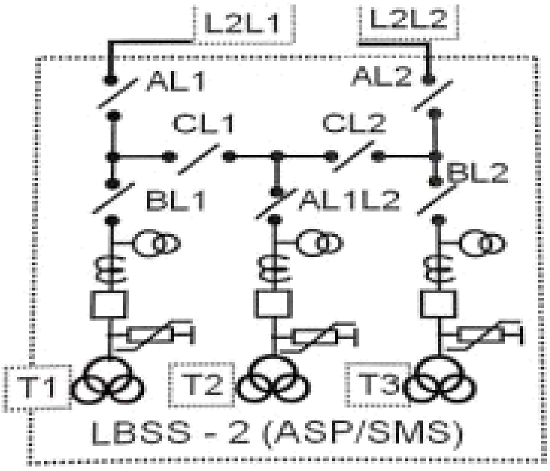
LBSS-4(LOAD BLOCK STEP-DOWN SUBSTATION -4):

This 220 KV Substation step-downs to 11 KV level and operates at a load of 4050MW.



LBSS-2(LOAD BLOCK STEP-DOWN SUBSTATION2):

At LBSS-2, THE 220KV power will be stepped down to 11kv and 6.6kv over three 220/11.5/6.9kv, 50/63/80 MVA, and three winding transformers for feeding various consumers over 11kv and 6.6 circuit breaker distribution board. The capacity of transformers selected is such that in the event of failure of one transformer, the other two will meet the entire demand of the substation. The LBSS-2 takes maximum load of approximately 70-80MW as it serves the main departments BF (BLAST FURNACE), SMS (STEEL MELTING SHOP), ASP(AIR SEPERATION PLANT),



LOAD BLOCK SUBSTATIONS:

LOAD BLOCK STEPDOWN SUBSTATION

- Load block stepdown substation receives power at 220kv and steps down to 33/11/6.6kv by transformers for further distribution of power in blocks to shops.
- Only at L.B.S.S 5(T.P.P) generator power (11kv) is stepped up to 220kv for power evacuation from generators through GSB.

GSB: Generator switch board

- Switch board connected to turbo generators and feeding category1 loads at 11 kv.
- It is linked to 220 kv grid through transformers.
- Power flow to/from grid depends on generator level.

LOAD BLOCK DISTRIBUTION SUBSTATION

- Receives power at 11/6.6 kv level
- Feeder multiplication to cater various loads of shops.

LOAD CENTRE STEPDOWN SUBSTATION

- Receives power at 11kv/6.6kv
- Steps down power to 440v through transformersDistributes power to LT loads.

HIGH VOLTAGE DISTRIBUTION (33/11/6.6 KV)

Two 220/33 KV Transformers installed at MRS feed power to Township step down station (called as CPRS) through 33 KV cables. Here the voltage is further stepped down to 11 KV by two nos. of 33/11 KV transformers. Out going feeders from this station supply power at 11 KV through cables to township net work. 11 KV overhead construction power lines are connected to this sub-station.The 33 KV supply form the transformer at LBSS2 feeds SMS ladle furnace transformer and capacitor banks through cables. This is a highly fluctuating load and the voltage dips on 220 KV systems can be felt when the furnace is in operation.Electric power at 11/6.6 KV stepped down at above LBSS stations is distributed to smaller Load Block Distribution Station (LBDS) located in each unit. Load Centre Sub-station (LCSS) transformers which convert power form11 KV to 415 V; converter, arc furnace, 11/6.6 KV High Voltage Load Centre (HVLC) transformers and 11 KV motors (above 200 kW rating generally) are connected to LBSS/LBDS, 11 KV switch boards through circuit breakers. Power supply to essential category loads of various zones is extended form TPP directly from Generator Switch Board (GSB) at 11 KV. Cross-linked Poly Ethylene (ELPE) cables are used for 11 KV distribution. 6.6 KV supply is mostly used for motors (of rating >200 and <200 kW generally). In the zones fed by LBSS3, LBSS4 stations and also where 11 KV supply from TPP is connected for motor drives, where ever required, step down 11/6.6 KV High Voltage Load Centre (HVLC) are formed with suitable capacity transformers (20/10/6.3 MVA capacity). High Voltage Motor Control Centre (HVMCC) is fed from 6.6 KV LBSS/LBDS/HVLC stations. 6.6 KV motors are connected through breaker to LBSS/LBDS/HVLC/HVMCC switch boards. Some times vacuum contactors are also used for motor switching. Cross-linked poly ethylene cables are used for 6.6 KV distribution.

directed to GSB - 2 and GSB-3 and the rest to LBSS-5. Some amount of the power from LBSS-5 is routed to LBSS - 1 and the surplus power which still exists is routed to MRS, which in turn exports to the APTRANSCO grid.

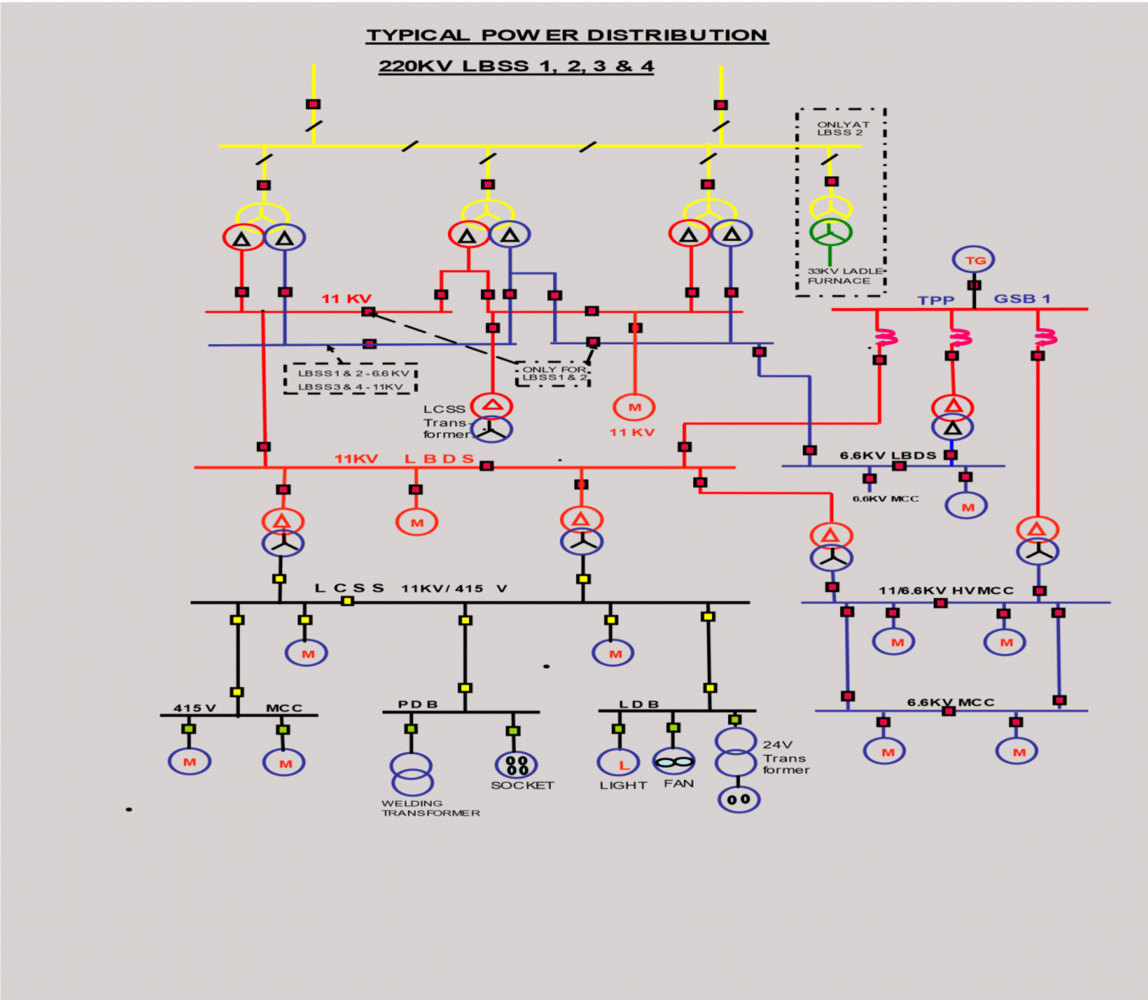


Fig: The above figure describes the distribution of power from load block substations to various loadsto limit the earth fault current to 1000/600 Amps respectively. Appropriate protective relays provided to quickly isolate faulty feeders/apparatus with suitable discrimination.

All varieties of breakers such as Bulk OIl/MOCB/VCB/SF_6/Air Blast are used.

MEDIUM VOLTAGE DISTRIBUTION (BELOW 650 V)

In each shop, to cater to Medium and Low voltage load, 11 KV/415 V LCSS are formed. Power is fed from LCSS to motors, MCC, Power Distribution Boards (PDB), Lighting Distribution Boards (LDB), ESP transformers etc. and for further distribution. The 3 phase 415 V distribution is solidly earthed neutral system. The transformers are of standard rating 1600/1000/750/630/250 KVA in plant and township areas. PVC cables are used for medium voltage distribution generally.

The secondary windings of converter transformers feed DC converters and other special equipment for variable speed AC/DC drives. The secondary voltage of these transformers suits to the particular application for which they are provided.

LOW VOLTAGE DISTRIBUTION (250 V & BELOW)

Low voltage distribution consists of power supply on single phase to lighting equipment, ceiling fans, portable hand tools and domestic appliances etc. For safety reasons, 24 V distribution is also provided to cater to hand lamps etc. in some areas.
PVC cables are used.

HIGH VOLTAGE LOAD CENTRE

Whenever direct 6.6 kv supply is not available ,11/6.6 kv transformers are provided for feeding HV motor loads.

HIGH VOLTAGE MOTOR CONTROL CENTRE

Directly receives power at 6.6 kv level

Feeders are multiplied to cater power to HV motors.

RELAYS

Relay Definition:

A relay is an electrical switch that opens and closes under the control of another electrical circuit. In the original form, the switch is operated by an electromagnet to open or close one or many sets of contacts. It was invented by Joseph Henry in 1835 HYPERLINK "http://encyclopedia.thefreedictionary.com/1835". Because a relay is able to control an output circuit of higher power than the input circuit, it can be considered, in a broad sense, to be a form of an electrical amplifier.

Operation:

When a current flows through the coil, the resulting magnetic field attracts an armature that is mechanically linked to a moving contact. The movement either makes or breaks a connection with a fixed contact. When the current to the coil is switched off, the armature is returned by a force approximately half as strong as the magnetic force to its relaxed position. Most relays are manufactured to operate quickly. In a low voltage application, this is to reduce noise. In a high voltage or high current application, this is to reduce arcing.

If the coil is energized with DC, a diode is frequently installed across the coil, to dissipate the energy from the collapsing magnetic field at deactivation, which would otherwise generate spike of voltage and might cause damage to circuit components. Some automotive relays already include that diode inside the relay case. Alternatively a contact protection network, consisting of a capacitor and resistor in series, may absorb the surge. If the coil is designed to be energized with AC, a small copper ring can be crimped to the end of the solenoid. This "shading ring" creates a small out-of-phase current, which increases the minimum pull on the armature during the AC cycle.By analogy with the functions of the original electromagnetic device, a solid- state relay is made with a thyristor or other solid-state switching device. To achieve electrical isolation an opto coupler can be used which is a light-emitting diode (LED) coupled with a phototransistor.

Protective relay:

A protective relay is a complex electromechanical apparatus, often with more than one coil, designed to calculate operating conditions on an electrical circuit and trip circuit breakers when a fault was found. Unlike switching type relays with fixed and usually ill- defined operating voltage thresholds and operating times, protective relays had well- established, selectable, time/current (or other operating parameter) curves. Such relays were very elaborate, using arrays of induction disks, shaded-pole magnets, operating and restraint coils, solenoid-type operators, telephone-relay style contacts, and phase-shifting networks

Over current relay:

An "Over current Relay" is a type of protective relay. The ANSI Device Designation Number is 50 for an Instantaneous over Current (IOC), 51 for a Time over Current (TOC). In a typical application the over current relay is used for over current protection, connected to a current transformer and calibrated to operate at or above a specific current level. When the relay operates, one or more contacts will operate and energize a trip coil in a Circuit Breaker and trip (open) the Circuit Breaker.

Applications:

- To control a high-voltage circuit with a low-voltage signal, as in some types of modems,
- To control a high-current circuit with a low-current signal, as in the starter solenoid of an automobile.

- To detect and isolate faults on transmission and distribution lines by opening and closing circuit breakers (protection relays),
- To isolate the controlling circuit from the controlled circuit when the two are at different potentials, for example when controlling a mains-powered device from a low-voltage switch. The latter is often applied to control office lighting as the low voltage wires are easily installed in partitions, which may be often moved as needs change. They may also be controlled by room occupancy detectors in an effort to conserve energy,

EARTH FAULT RELAY: It is a safety device used in electrical installations with high earth impedance. It detects small stray voltages on the metal enclosures of electrical equipment. The result is to interrupt the circuit if a dangerous voltage is detected. The EFR is protected against tripping from transients and prevents shock. The devices give the tripping command to break the circuit when earth fault occurs. The fault current is restricted and the fault is dispersed by the Restricted Earth Fault Protection (REFP) scheme. Normally earth fault relay, earth leakage circuit breaker and ground fault circuit interrupter, etc. are used to restrict the fault current. Earth Fault is an inadvertent fault between the live conductor and the earth. When earth fault occurs, the electrical system gets short-circuited and the short-circuited current flows through the system. The fault current returns through the earth or any electrical equipment.

IDMT (Inverse Definite Minimum Time) Relay:-

Operates after a time delay that decreases as the fault current increases- Trip time is inversely proportional to the fault current magnitude-

Example: – Setting: IDMT relay set to trip at 100ms for any current above 500A – Fault current: 550A – Trip time: 100ms – Fault current: 750A – Trip time: 50ms (decreases as fault current increases)

In summary DMT relay trips after a fixed time delay, regardless of fault current magnitude-IDMT relay trips after a time delay that decreases as the fault current increases

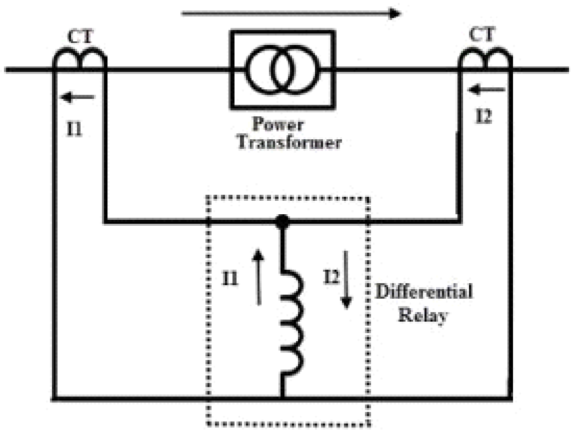
These relays are used in electrical power systems to protect against overcurrent conditions, such as short circuits and overload.

The choice between DMT and IDMT depends on the specific protection requirements and system characteristics.

DIFFERENTIAL RELAY:

The relay which works once the phasor difference for a minimum of two or above the same electrical quantities exceeds a fixed amount is known as a differential relay. Generally, most of the relays work when any quantity goes beyond a fixed value however, this relay works based on the difference between two or more same electrical quantities.

The function of a differential relay is to provide high-speed, sensitive & naturally selective protection. These relays will not provide safety to turn-to-turn winding faults within machines and transformers due to the small growth within the generated current by those faults, which stay under the relay's pickup sensitivity.



In normal operating conditions, the entering & leaving currents are equivalent in phase & magnitude thus the relay will not work. However, if any fault occurs within the system, then these flow of currents will be no longer equivalent in phase & magnitude. This kind of relay is used in such a way that the difference between the entering & leaving current supplies throughout the relay's operating coils. So, the relay coil can be energized in fault conditions because of the various quantity of the current. So, this relay functions & opens the circuit breaker for tripping the circuit.

In the above differential relay circuit, there are two current transformers which are connected to any face of the power transformer like one CT is connected on the primary side and the other is connected at the secondary side of the PT (power transformer). This relay simply compares the flow of currents on both sides. If there is any unbalance in the current flow of the circuit then this relay tends to operate. These relays can be current differential, voltage balance & biased **Types of faults in Transmission Lines and Transformers:**

- Earth fault
- Over current fault

Types of Protection:

- Primary Protection
- Back up Protection or Secondary Protection

Primary Protection should work instantaneously after fault occurs. Secondary Protection is for reliable purpose if primary protection fails.

Differential Protection is known as Primary Protection. Generally in VSP pilot wire protection is used (short length < 15 km)

Back up is earth fault and over current relay protection.

Relays :

<u>For Incomer</u>

- IDMT
- Over Current Relay
- Trip Circuit Supervision Relay
- Master Trip Relay

<u>Line PT</u>

- Fuse Failure Relay
- Over current Relay

<u>Outgoing Feeder</u>

- IDMT
- Instantaneous Over Current Relay
- Earth Fault Relay
- Trip Relay for Transformer fault
- Master Trip Relay
- Trip Circuit Supervision Relay
- Time Delay for Transformer Earth fault Relay

<u>Bus PT</u>

- Fuse Failure Relay

<u>Bus Coupler</u>

- Earth Fault Relay
- Master Trip Relay
- Trip Circuit Supervision Relay

<u>Transformer Protection</u>

- Fuse fail relay

- IDMT O/C relay
- Trip relay
- Trip Circuit Supervision relay
- Auxiliary relay
- Master Trip Relay:
- If any relay operates during the fault occurrence then the master trip relay operates.
- Trip Circuit Supervision Relay:
- To Ensure the healthiness of the relays this relay is used.
- For any Protection scheme these Master Trip Relay and Trip Circuit Supervision Relay are used

CONCLUSION

During this project work, we have visited the various sections namely, LBSS- 4 in DNW department and got acquaintance with various equipment like 220KV transformers, circuit breakers, CT's, PT's, lightning arrestors, isolators. We also came to know the operational procedures followed during various conditions. Through this project we all felt this is extremely helpful for improving the technical knowledge.

Power distribution plays a major role in the development of industrial sector in current generation. The various equipment installed for the distribution plays a major role which includes transmission lines, 220 KV lighting arrestors, current transformers, potential transformers, isolators, circuit breakers, transformers, ACSR conductors, insulators structures, Relay and control panels, batteries etc.

The power distribution scheme in VSP has been designed in such a way that continuity and reliability of the system is more and all the units of plant get continuous supply.